Extracting Market-Implied Federal Reserve Policy Rate Probabilities from 30-Day Federal Funds Futures: Theory, Identification, and Implementation

Angelo Righetti

Università degli Studi di Trento · angelo.righetti@studenti.unitn.it

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Abstract

This research presents the theoretical and operational foundations of a framework that extracts market-implied probabilities of Federal Reserve policy rate decisions (hikes, holds, and cuts) at upcoming FOMC meetings from the settlement prices of 30-Day Federal Funds futures. After describing the institutional setting (the target range, the Effective Federal Funds Rate, and the administered rates of the floor system), I derive the day-weighted extraction equation that isolates the expected post-meeting rate within a contract month, characterize how measurement error is amplified for meetings held late in the month, and show that a next-month rule resolves the problem. Under a zero term-premium assumption at short horizons, the expected policy move is mapped into a two-point distribution over adjacent 25-basis-point steps, which I prove to be the minimum-variance distribution consistent with the futures-implied mean. Multi-meeting rate paths are obtained by discrete convolution. The paper closes with an explicit discussion of the assumptions and limitations of the approach.

Keywords: Federal funds futures; FOMC; monetary policy expectations; implied probabilities; term premium; policy rate path

JEL Classification: E43, E52, E58, G13

1 Introduction

The path of the policy rate set by the Federal Open Market Committee (FOMC) is among the most closely monitored variables in financial markets. The allocation of capital across bonds, equities, and currencies, the cost of corporate and household credit, and the valuation of virtually every long-lived asset depend on the expected trajectory of short-term interest rates (Bernanke and Kuttner 2005). Because the Federal Reserve moves in discrete increments at scheduled meetings, the relevant object for market participants is not only the expected level of the policy rate but also its probability distribution across upcoming decisions: the likelihood of a hike, a cut, or no change.

Federal funds futures are the natural source of this information. Krueger and Kuttner (1996) show that the federal funds futures rate is an efficient predictor of Federal Reserve policy, Kuttner (2001) uses the same contracts to separate expected from unexpected policy actions, and Gürkaynak, Sack, and Swanson (2007) document that market-based measures of policy expectations are accurate at short horizons, where risk premia are small. At longer horizons, Piazzesi and Swanson (2008) show that futures rates embed time-varying risk premia that must be taken into account. Practitioner tools such as the CME FedWatch tool (CME Group n.d.b) and open methodologies such as RateProbability (n.d.) operationalize these ideas to produce meeting-by-meeting probabilities.

This paper provides a self-contained derivation of such a framework, designed to be implemented transparently and replicated. It makes four contributions. First, it derives the day-weighted extraction equation that recovers the expected post-meeting rate from a single monthly contract and clarifies the role of the time-weighting multiplier. Second, it quantifies how measurement error is amplified for meetings late in the month and proposes a next-month rule that removes the problem. Third, it proves that the two-point distribution over adjacent policy steps is the minimum-variance distribution consistent with the futures-implied mean, and so makes explicit what the resulting probabilities do and do not identify. Fourth, it chains meetings through discrete convolution to obtain the distribution of the target range along the policy horizon. An implementation as an interactive dashboard, built on data from FRED and CME settlement prices, is described at the conceptual level.

The remainder of the paper is organized as follows. Section 2 reviews the institutional background. Section 3 describes the 30-Day Federal Funds futures contract. Section 4 presents the identification of the expected policy rate. Section 5 maps expected moves into probability

distributions. Section 6 discusses assumptions and limitations, and Section 7 concludes. Appendix A provides a numerical illustration.

2 Institutional Background

2.1 The Federal Reserve System and its Mandate

The Federal Reserve System is the central bank of the United States. It was created in 1913 to provide the nation with a safer, more flexible, and more stable monetary and financial system (Board of Governors of the Federal Reserve System n.d.b). It is composed of three key entities:

1. **The Board of Governors** is the governing body of the Federal Reserve System. It is run by seven members, or "governors," who are nominated by the President of the United States and confirmed by the Senate.
2. **The twelve Federal Reserve Banks**, together with their 24 branches, are the operating arms of the System. Each Reserve Bank operates within its own geographic district, where it gathers data and information about local businesses and communities. This information feeds into the monetary policy decisions of the FOMC and into other decisions of the Board of Governors. Reserve Banks are not operated in the interest of shareholders but in the public interest.
3. **The Federal Open Market Committee (FOMC)** is the body that sets national monetary policy. It consists of the seven members of the Board of Governors, the president of the Federal Reserve Bank of New York, and four of the remaining eleven Reserve Bank presidents, who serve one-year terms on a rotating basis. The Chair of the Board is the FOMC Chair.

In 1977 Congress amended the Federal Reserve Act to specify the objectives that give rise to the so-called "dual mandate": maximum employment and price stability.[1] This mandate ties monetary policy to the broader goal of fostering a productive and stable U.S. economy.

2.2 Monetary Policy and its Transmission

Monetary policy comprises the actions taken by the Federal Reserve to achieve its statutory mandate. It operates primarily by influencing the cost and availability of money and credit

[1] The statutory text also lists "moderate long-term interest rates" as an objective. These are generally regarded as following from the achievement of the other two goals, which is why the objectives are commonly referred to as a dual mandate.

throughout the economy. Although policy can be implemented through several tools (discount window lending, open market operations, and balance sheet policies), the most important instrument is the target range for the federal funds rate (Board of Governors of the Federal Reserve System n.d.a).

Monetary policy most directly affects the current and expected future path of short-term interest rates. This anticipated path, in turn, affects overall financial conditions, including longer-term interest rates, stock prices, the exchange value of the dollar, and many other asset prices. Through these channels, policy influences the spending and investment decisions of households and businesses, and thereby production, employment, and inflation (Figure 1).

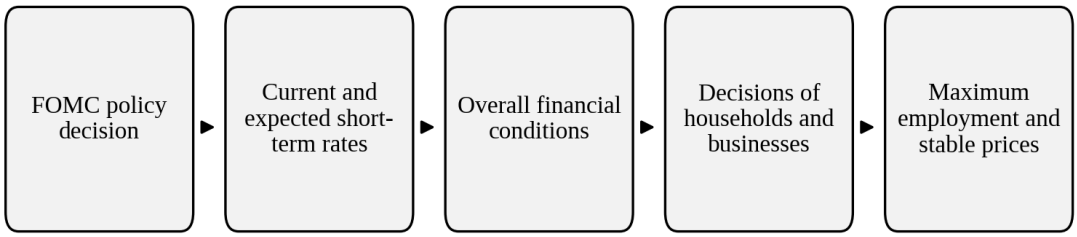


Figure 1. *Transmission of FOMC monetary policy decisions to the real economy.*

The initial impulse of the transmission chain is a single decision: the calibration of short-term interest rates. By changing the baseline cost of borrowing, the Federal Reserve alters the intertemporal consumption decisions of households, the hurdle rates of corporate investment, and overall credit creation. Rather than dictating loan rates directly, however, the central bank intervenes at the wholesale foundation of the financial system: the market for overnight interbank liquidity.

2.3 The Overnight Reserve Market and the Target Range

Commercial banks hold reserve balances in master accounts at the Federal Reserve. These balances are not commercial deposits; they are the risk-free settlement medium of the entire payments system. Every business day, a very large volume of customer payments, interbank settlements, and securities transactions clears across Fedwire. Because these gross flows cannot be perfectly synchronized, depository institutions end each day with structural imbalances. Banks in deficit face outflows in excess of inflows and must borrow reserves before the clearing window closes, in order to satisfy prudential requirements (such as the Liquidity Coverage Ratio) and to avoid overdraft charges at the central bank. Banks in surplus hold balances that carry an opportunity cost if left idle. Since a bank's liquidity position can reverse by the next

morning, these imbalances are resolved at an overnight tenor in the *federal funds market*, where depository institutions negotiate uncollateralized reserve transfers.

The FOMC does not set a single administered rate for these private transactions. Instead, it establishes a 25-basis-point *target range* (for example, 5.00%–5.25%) to steer the Effective Federal Funds Rate (EFFR). The EFFR is the volume-weighted median of unsecured overnight loans of reserve balances between depository institutions. Because counterparties differ in their intraday liquidity needs, bilateral credit assessments, and balance sheet constraints, individual transactions clear with modest dispersion, and the target range acts as an operational tolerance band that accommodates these frictions.

2.4 From Corridor to Floor System

Before 2008, aggregate reserves were scarce and reserve demand was elastic. The Federal Reserve steered rates by adjusting the daily quantity of reserves through open market operations. The corridor was bounded from above by the discount rate and from below by zero, because reserves earned no interest. After 2008, large-scale asset purchases (so-called Quantitative Easing) flooded the banking system with reserves and moved supply onto the flat segment of the demand curve, where quantity adjustments became ineffective. To retain control of short-term rates, the Federal Reserve began paying interest on reserve balances, which transitioned the framework to an administered-price *floor system* (Ihrig, Meade, and Weinbach 2015).

Within this framework, the EFFR is kept inside the target range by a set of administered rates:

- **Interest on reserve balances (IORB)**, set near the top of the target range, is the risk-free rate paid by the Federal Reserve to depository institutions. It determines their minimum reservation rate for lending reserves in the unsecured market.
- **The overnight reverse repurchase agreement (ON RRP) rate**, set at the bottom of the target range, is offered to nonbank cash lenders, such as money market mutual funds and government-sponsored enterprises (GSEs), which cannot earn IORB. Since these counterparties can always lend to the Federal Reserve at this rate, it establishes a floor for money market rates.
- **The ceiling.** The primary credit (discount) rate is set at the upper bound of the target range, and the Standing Repo Facility (SRF) offers overnight repurchase agreements against Treasury and agency collateral at a minimum bid rate also set at the upper

bound. These facilities limit upward pressure on overnight rates. Use of the discount window continues to carry a supervisory stigma, as borrowing there may signal liquidity stress, which discourages routine use.

In equilibrium, GSEs lend overnight cash to commercial banks at a slight discount to IORB. The banks borrow these funds and redeposit them at the Federal Reserve to capture the spread. This arbitrage compresses market rates and anchors the EFFR between the ON RRP floor and IORB. Typically,

$$\text{ON RRP} \lesssim \text{EFFR} \lesssim \text{IORB} \tag{1}$$

where the EFFR trades a few basis points below IORB and may approach or exceed it when reserves become less ample. When the FOMC moves these administered rates in tandem, the shift propagates across the entire credit and money market spectrum.

It is therefore essential for investors, businesses, and households to assess where the Federal Reserve is headed. Decisions such as allocating capital across asset classes, financing investment with debt, or taking out a mortgage or student loan are all linked to the path of the policy rate. A standard market-based benchmark for this purpose is the *market-implied policy rate probability*, which uses the prices of interest rate derivatives to infer where the market expects the next Federal Reserve decisions to land.

3 The 30-Day Federal Funds Futures Contract

3.1 Market-Implied Policy Rate Probabilities

Market-implied policy rate probabilities are the discrete probability distributions over future policy rate adjustments (hikes, holds, or cuts), derived from the prices of interest rate derivatives and money-market instruments such as Overnight Index Swaps (OIS) and 30-Day Federal Funds futures and options.

3.2 Contract Specification

The 30-Day Federal Funds futures contract (CME ticker ZQ) is a standardized, cash-settled contract that pays according to the average level of the overnight federal funds rate over a calendar month (CME Group n.d.a). No principal changes hands. The contract serves two main purposes. Commercial banks and other borrowers use it to hedge: a bank expecting to borrow large amounts of overnight reserves in November faces the risk of a hike at its next meeting,

and by selling November futures it obtains a gain on the exchange that offsets higher physical borrowing costs, since a rate increase depresses the contract price. Macro desks and asset managers use it to speculate: buying contracts expresses a view that rates will fall, and selling contracts expresses a view that they will rise.

Contracts are listed for 60 consecutive months and settle in cash on the last business day of the month. The contract price is 100 minus the arithmetic average of the daily EFFR over the contract month:

$$P_m = 100 - R_{\text{avg},m} \tag{2}$$

so that the average monthly rate implied by the settlement price is

$$R_{\text{avg},m} = 100 - P_m = \frac{1}{N_m} \sum_{d=1}^{N_m} \text{EFFR}_{m,d} \tag{3}$$

where $\text{EFFR}_{m,d}$ is the EFFR published by the Federal Reserve Bank of New York for day d of month m, and $N_m \in \{28,29,30,31\}$ is the number of calendar days of the contract month. Despite the name "30-Day," the contract thus refers to the calendar days of the month and not to a fixed 30-day horizon. The quotation $100 - R_{\text{avg},m}$ replicates the inverse price–rate relationship typical of fixed-income securities: a decline in rates raises the futures price, so long positions profit from cuts and short positions profit from hikes.

The New York Fed publishes the EFFR only for business days, whereas the sum in Eq. (3) runs over all calendar days of the month. By CME rule, the rate of each non-business day equals the rate of the preceding business day. A Friday rate therefore carries a weight of three days in the monthly average (Friday, Saturday, and Sunday), and four days when the following Monday is a federal holiday.

3.3 Marginal Measure: the DV01

The DV01 (dollar value of a basis point) is the absolute change in the contract value caused by a one-basis-point change in the underlying rate. The contract has a notional value of $5,000,000 under money-market conventions (ACT/360 over a 30-day baseline):

$$\text{DV01} = 5{,}000{,}000 \times 0.0001 \times \frac{30}{360} = \$41.67 \tag{4}$$

The notional was chosen to align the rate sensitivity of the contract with other short-term interest rate derivatives. A standard three-month contract with a $1,000,000 face value over 90 days has a DV01 of exactly $25.00. A $1,000,000 Fed Funds contract covering only 30 days would have a DV01 of only $8.33, which would force institutional desks to trade excessive volumes (hedging a $50,000 per basis point exposure would require 6,000 contracts instead of 1,200). The $5,000,000 notional yields a DV01 of $41.67 and, for the front-month contract, a minimum price increment of 0.0025 (one quarter of a basis point), equal to $10.42 per tick. Deferred months trade in increments of 0.005, equal to $20.83.

4 Identification of the Expected Policy Rate

4.1 Term Premium

Let $\mathbb{Q}$ denote the risk-neutral measure and $\mathbb{P}$ the physical probability measure. Neglecting margining and convexity effects, the futures rate is the risk-neutral expectation of the monthly average EFFR. Following Piazzesi and Swanson (2008), it can be decomposed as

$$R_{\text{avg},m}^{(t)} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{1}{N_m}\sum_{d=1}^{N_m}\text{EFFR}_{m,d}\right] = \mathbb{E}_t^{\mathbb{P}}\left[\frac{1}{N_m}\sum_{d=1}^{N_m}\text{EFFR}_{m,d}\right] + \rho_{t,m} \tag{5}$$

where $\rho_{t,m}$ is the term premium. Reading policy expectations directly from $R_{\text{avg},m} = 100 - P_m$, that is, equating the futures rate with the conditional expectation under $\mathbb{P}$, implicitly assumes $\rho_{t,m} = 0$.

This assumption is empirically reasonable for front-month contracts. Krueger and Kuttner (1996) and Gürkaynak, Sack, and Swanson (2007) find that term premia at short horizons (one to three months) are small and statistically indistinguishable from zero, on the order of a few basis points. At longer horizons (six to twelve months), premia become economically meaningful and volatile, reflecting the compensation that market participants demand for monetary policy uncertainty over the business cycle (Piazzesi and Swanson 2008). The framework below adopts the zero-premium assumption throughout, and Section 7 discusses the consequences for distant meetings.

4.2 The FOMC Calendar

By statute, the FOMC must meet at least four times each year. Since 1981, eight regularly scheduled meetings have been held annually, at intervals of five to eight weeks. Table 1 shows the 2026 schedule together with the identification diagnostics developed in the following subsections. The policy statement is released at 2:00 p.m. Eastern Time (ET) on the second day of each two-day meeting.

Table 1. *FOMC meetings in 2026 and identification diagnostics.*

#	Meeting	M_m	N_m	$N_m - M_m$	$N_m/(N_m - M_m)$	Estimator
1	January 27–28	28	31	3	10.33	Next-month (Feb.)
2	March 17–18*	18	31	13	2.38	Own-month
3	April 28–29	29	30	1	30.00	Next-month (May)
4	June 16–17*	17	30	13	2.31	Own-month
5	July 28–29	29	31	2	15.50	Next-month (Aug.)
6	September 15–16*	16	30	14	2.14	Own-month
7	October 27–28	28	31	3	10.33	Next-month (Nov.)
8	December 8–9*	9	31	22	1.41	Own-month

Notes: M_m is the day of the month on which the decision is announced; N_m is the number of calendar days of the month. Meetings marked with an asterisk are accompanied by the Summary of Economic Projections. The estimator column applies the rule of Section 4.4 with threshold $L = 7$ days.
Source: Board of Governors of the Federal Reserve System.

4.3 Day-Weighted Extraction of the Post-Meeting Rate

Consider a month m with N_m calendar days. Suppose that a scheduled FOMC meeting concludes on day M_m and that the policy statement is released that afternoon. If the FOMC changes the target range on day M_m, the new rate takes effect on the following business day, $M_m + 1$. The prevailing (old) rate therefore applies to the whole of day M_m, and the month divides into two segments:

1. a *pre-meeting segment* of M_m days (day 1 to day M_m, inclusive), during which the rate is the pre-meeting rate $R_{\mathrm{pre},m}$;
2. a *post-meeting segment* of $N_m - M_m$ days (day $M_m + 1$ to the end of the month), during which the revised rate $R_{\mathrm{post},m}$ applies.

Assuming that the EFFR is constant within each segment (that is, abstracting from idiosyncratic liquidity noise), the monthly average becomes a time-weighted average:

$$R_{\mathrm{avg},m} = \left(\frac{M_m}{N_m}\right) R_{\mathrm{pre},m} + \left(\frac{N_m - M_m}{N_m}\right) R_{\mathrm{post},m} \tag{6}$$

The settlement price of the contract for month m gives $R_{\text{avg},m} = 100 - P_m$. If the pre-meeting rate is known, the only unknown in Eq. (6) is the expected post-meeting rate. Multiplying both sides by N_m and solving for $R_{\text{post},m}$ yields the *extraction equation*:

$$R_{\text{post},m} = \frac{N_m\, R_{\text{avg},m} - M_m\, R_{\text{pre},m}}{N_m - M_m} \tag{7}$$

Equivalently, in terms of deviations from the prevailing pre-meeting rate,

$$R_{\text{post},m} - R_{\text{pre},m} = \left(\frac{N_m}{N_m - M_m}\right)\left(R_{\text{avg},m} - R_{\text{pre},m}\right) \tag{8}$$

The expected policy shift is therefore the difference between the market-implied monthly average and the pre-meeting rate, scaled by the ***time-weighting multiplier*** $N_m/(N_m - M_m)$. This day-weighting is the same algebra that Kuttner (2001) applies to changes in the current-month futures rate on announcement days in order to identify the unexpected component of policy; Eq. (8) applies it to levels in order to recover the expected component.

4.4 Error Amplification and the Next-Month Rule

Because the multiplier in Eq. (8) scales the entire right-hand side, any measurement error $\varepsilon_{\text{avg},m}$ in the implied average rate (arising from tick rounding, bid–ask noise, end-of-month EFFR dispersion, or a violation of the constant-rate assumption) is amplified by the same factor:

$$\varepsilon_{\text{post},m} = \frac{N_m}{N_m - M_m}\, \varepsilon_{\text{avg},m} \tag{9}$$

When the policy step is $\delta = 0.25$ percentage points, the sensitivity of the implied probability of the larger step to the futures-implied rate is

$$\frac{\partial\, \alpha_m}{\partial\, R_{\text{avg},m}} = \frac{N_m}{(N_m - M_m)\, \delta} \tag{10}$$

where α_m is defined in Section 5. For a meeting held on October 28 ($N_m = 31$, $M_m = 28$), the multiplier is 10.33: an error of one basis point in $R_{\text{avg},m}$ translates into an error of 10.3 basis points in the expected post-meeting rate and of 41 percentage points in the implied probability.

For the April 29 meeting ($N_m = 30$, $M_m = 29$), the multiplier is 30, so that the same error exceeds a full policy step. Figure 2 shows the multiplier as a function of the decision day, and Table 1 reports it for each 2026 meeting.

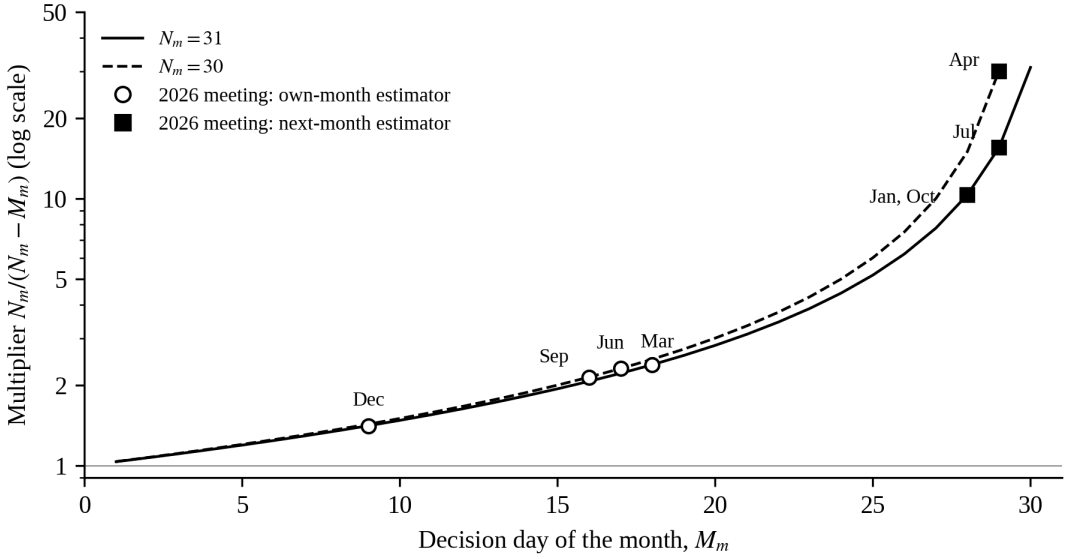


Figure 2. *Time-weighting multiplier $N_m/(N_m - M_m)$ as a function of the decision day M_m. Squares denote 2026 meetings for which the next-month estimator is used, and circles denote meetings for which the own-month estimator is used.*

The remedy is to read the post-meeting rate from the contract of the *following* month whenever that month contains no meeting. In a month without an FOMC decision, the policy rate is constant, so the price of its contract equals the expected rate prevailing after the previous meeting; this is the logic exploited by the CME FedWatch methodology, which chains rate expectations across months with and without meetings (CME Group n.d.b). Formally, if $N_m - M_m < L$ and month $m + 1$ has no scheduled meeting, then

$$R_{\text{post},m} = R_{\text{avg},m+1} = 100 - P_{m+1} \tag{11}$$

The threshold L is a design parameter. This research uses $L = 7$ days, so that the own-month multiplier never exceeds $N_m/7 \approx 4.4$. In the 2026 calendar the rule applies to the January, April, July, and October meetings, and in each case the following month (February, May, August, November) contains no meeting, so the condition is always satisfied (Table 1). If month $m + 1$ did contain a meeting, Eq. (11) would not hold, because the next-month contract would mix two unknown post-meeting rates. In that case the own-month formula, Eq. (7), is used for month m, the resulting estimate is flagged as low reliability, and the contract of month $m + 1$ is then used to solve for $R_{\text{post},m+1}$ sequentially. The extraction equation is undefined

when $N_m = M_m$ (a meeting on the last day of the month), in which case the next-month rule always applies.

4.5 The Current Month

If the evaluation date falls inside month m, a part of the monthly average is already realized. Let t be the number of days already fixed, with realized sum $S_t = \sum_{d=1}^{t} \mathrm{EFFR}_{m,d}$, and suppose that $t < M_m$. Only the remaining days are then unknown, and the extraction equation becomes

$$R_{\mathrm{post},m} = \frac{N_m\, R_{\mathrm{avg},m} - S_t - (M_m - t)\, R_{\mathrm{pre},m}}{N_m - M_m} \tag{12}$$

If $t \geq M_m$, the meeting has already taken place and the contract for month m contains no information on future decisions; the first unresolved meeting is then identified from subsequent contracts.

4.6 The Policy Rate and the EFFR

The EFFR is not the policy rate: it is an outcome of the market, and it trades at a spread above the lower bound of the target range, a spread which varies with money market conditions (for example, month-end and quarter-end repo pressure, or changes in reserve supply). Writing $\mathrm{EFFR}_d = L_d + s_d$, where L_d is the lower bound of the target range and s_d the EFFR–floor spread, the first pre-meeting rate can be anchored at

$$R_{\mathrm{pre},1} = L_0 + \hat{s} \qquad \hat{s} = \mathrm{median}_{\tau \in \mathcal{T}_{20}}(\mathrm{EFFR}_\tau - L_\tau) \tag{13}$$

where $\mathcal{T}_{20}$ denotes the last 20 business days. The spread is assumed to be constant over the horizon and unaffected by FOMC decisions, so that a policy step of size Δ moves both L and the EFFR by Δ. Using the median instead of the last print reduces the sensitivity of the anchor to transitory dispersion. Changes in s_d are otherwise misread as changes in policy expectations, which is a limitation of the method (Section 7).

5 From an Expected Move to a Probability Distribution

5.1 The Two-Point Distribution

Eq. (7) delivers a single number: the expected post-meeting rate, for example 3.80%. To obtain a probability distribution, the expected move in units of standard policy steps needs to be expressed. Let $\delta = 0.25$ percentage points and

$$H_m = \frac{R_{\text{post},m} - R_{\text{pre},m}}{\delta} \tag{14}$$

so that $H_m > 0$ indicates that the market prices a hiking regime and $H_m < 0$ a cutting regime. Decompose its absolute value into an integer part (the *characteristic* k_m) and a fractional part (the *mantissa* α_m), and let s_m denote the direction of the move:

$$k_m = \lfloor |H_m| \rfloor \in \{0,1,2,\dots\}, \qquad \alpha_m = |H_m| - k_m \in [0,1), \qquad s_m = \operatorname{sign}(H_m) \in \{-1,+1\} \tag{15}$$

The model allows only two outcomes at each meeting: a *lower step* $\Delta_{\text{low},m} = s_m k_m \delta$ with probability p_{low}, and an *upper step* $\Delta_{\text{high},m} = s_m (k_m + 1)\delta$ with probability p_{high}.

To illustrate, consider an upcoming meeting m where the current policy rate is $R_{\text{pre},m} = 3.50\%$ and the market prices an expected post-meeting rate of $R_{\text{pre},m} = 3.65\%$ with a standard policy increment $\delta = 0.25$. From Eq. (14), the expected rate change in standard step units is:

$$H_m = \frac{3.65\% - 3.50\%}{0.25\%} = 0.60$$

Since $H_m > 0$, the market expects a rate hike, yielding $s_m = +1$. Decomposing $|H_m| = 0.60$ according to Eq. (15) gives the integer characteristic $k_m = \lfloor 0.60 \rfloor = 0$ and the mantissa $alpha_m = 0.60 - 0 = 0.60$. Consequently, the two admissible adjacent outcomes are:

- A lower step: $\Delta_{\text{low},m} = (+1) \times 0 \times 0.25 = 0{,}00\%$ (unchanged rate) with probability $p_{\text{low}} = 1 - \alpha_m = 0.40$.
- An upper step: $\Delta_{\text{high},m} = (+1) \times (0 + 1) \times 0.25 = +0{,}25\%$ (a 25 bps hike) with probability $p_{\text{high}} = \alpha_m = 0.60$.

The resulting discrete distribution satisfies $\mathbb{E}[\Delta_m] = 0.40 \times 0.00\% + 0.60 \times 0.25\% = 0.15\%$, matching the market expectation of $R_{\text{post},m} = 3.65$.

Probabilities must sum to one, and the expected move, measured in steps, must equal $|H_m|$:

$$\begin{aligned} p_{\text{low}} + p_{\text{high}} &= 1, \\ p_{\text{low}}\, k_m + p_{\text{high}}\, (k_m + 1) &= k_m + \alpha_m \end{aligned} \tag{16}$$

Substituting $p_{\text{low}} = 1 - p_{\text{high}}$ into the second equation gives $k_m + p_{\text{high}} = k_m + \alpha_m$, hence

$$p_{\text{high}} = \alpha_m \qquad p_{\text{low}} = 1 - \alpha_m \tag{17}$$

The mantissa is therefore identically equal to the probability of the larger policy increment, and the residual mass falls on the adjacent smaller step.

Returning to our earlier example with $R_{\text{post},m} - R_{\text{pre},m} = +0{,}15$, where $H_m = 0.60$, $k_m = 0$, $\alpha_m = 0.60$, the model assigns a 60% probability to a 25 bp hike and 40% to no change.

A single Fed Funds futures price identifies only the expected policy rate after an FOMC meeting, leaving the underlying probability distribution underdetermined. The two-point specification resolves this identification issue by restricting probability mass exclusively to the two grid steps immediately adjacent to the market mean, strictly bounding the support within the bracket $[\Delta_{\text{low},m}, \Delta_{\text{high},m}]$ (e.g., a pause and a single hike when $s_m = +1$). While infinitely many discrete combinations, such as offsetting rate cuts with outsized hikes, could match the same expected value, assigning weight to policy moves contrary to s_m would mechanically widen the support and inflate the distribution's variance. By confining outcomes strictly to adjacent states, this setup yields the most parsimonious distribution possible and uniquely minimizes dispersion around the market-implied mean.

5.2 Forward Chaining

To evaluate expectations over a sequence of upcoming meetings, the model proceeds chronologically through the future months $m = 1,2,\ldots,J$. The *initial state* is the pre-meeting rate $R_{\text{pre},1}$ of Eq. (13). For each subsequent month:

- **Months without a meeting (anchor months).** No policy change occurs, and the market price provides an empirical recalibration of the rate carried forward:

$$R_{\text{pre},m+1} = R_{\text{avg},m} = 100 - P_m \tag{19}$$

- **Meeting months.** The post-meeting rate $R_{\text{post},m}$ is computed with the estimator given in Table 1 (Eq. (7), Eq. (11), or Eq. (12)), and the subsequent period inherits it:

$$R_{\text{pre},m+1} = R_{\text{post},m} \tag{20}$$

The expected cumulative policy adjustment from current conditions through meeting j, and the number of standard steps priced in, are

$$\Delta R_{\mathrm{cum},j} = R_{\mathrm{post},j} - R_{\mathrm{pre},1} \qquad \text{Priced-in steps}_j = \frac{\Delta R_{\mathrm{cum},j}}{\delta} \tag{21}$$

5.3 Multi-Meeting Convolution

Each meeting j defines a discrete distribution over policy adjustments, summarized by the vector

$$q_j = \begin{pmatrix} \Delta_{\mathrm{low},j} & \Delta_{\mathrm{high},j} \\ 1-\alpha_j & \alpha_j \end{pmatrix} \tag{22}$$

whose first row lists the possible adjustments and whose second row lists their probabilities. Let $P_j(\ell)$ denote the probability mass of the lower bound of the target range being equal to ℓ after meeting j. Starting from $P_0(L_0) = 1$, the cumulative distribution is updated by discrete convolution:

$$\begin{aligned} P_j &= P_{j-1} * q_j \qquad P_j(\ell) \\ &= (1-\alpha_j)\, P_{j-1}(\ell - \Delta_{\mathrm{low},j}) + \alpha_j\, P_{j-1}(\ell - \Delta_{\mathrm{high},j}) \end{aligned} \tag{23}$$

Eq. (23) is the convolution of independent two-point increments, equivalent to a Chapman–Kolmogorov step for a random walk on the policy grid. It generates a complete matrix of meeting-by-meeting probability buckets across the policy horizon.

By linearity of expectations, the mean of the cumulative distribution reproduces the expected path implied by the futures:

$$\sum_{\ell} \ell\ P_j(\ell) = L_0 + \delta \sum_{i=1}^{j} H_i, \tag{24}$$

provided that the pre-meeting rate of each meeting equals the post-meeting rate of the previous one, which is guaranteed by Eqs. (19) and (20) in the absence of reconciliation gaps (Section 7). The marginal probabilities of a hike, a cut, or no change at meeting j are read directly from q_j.

To illustrate the discrete convolution across meetings, consider a horizon of two consecutive meetings ($j = 1,2$) starting from an initial policy rate $L_0 = 3.50\%$. Suppose the first meeting yields policy adjustment vector:

$$\mathbf{q}_1 = \begin{pmatrix} 0.00\% & +0.25\% \\ 0.40 & 0.60 \end{pmatrix}$$

resulting in post-meeting distribution $P_1(3.50\%) = 0.40$ and $P_1(3.75\%) = 0.60$. For the second meeting, let the marginal distribution be:

$$\mathbf{q}_2 = \begin{pmatrix} 0.00\% & +0.25\% \\ 0.30 & 0.70 \end{pmatrix}$$

Applying Eq. (23), the cumulative probability distribution after the second meeting, $P_2 = P_1 * q_2$, is obtained by convolving the two steps:

$$\begin{aligned} P_2(3.50\%) &= P_1(3.50\%) \times (1 - \alpha_2) = 0.40 \times 0.30 = 0.12, \\ P_2(3.75\%) &= P_1(3.75\%) \times (1 - \alpha_2) + P_1(3.50\%) \times \alpha_2 \\ &= (0.60 \times 0.30) + (0.40 \times 0.70) = 0.18 + 0.28 = 0.46, \\ P_2(4.00\%) &= P_1(3.75\%) \times \alpha_2 = 0.60 \times 0.70 = 0.42. \end{aligned}$$

The probabilities sum to unity $(0.12 + 0.46 + 0.42 = 1.00)$, and the expected policy rate satisfies Eq. (24):

$$\mathbb{E}[L_2] = (0.12 \times 3.50\%) + (0.46 \times 3.75\%) + (0.42 \times 4.00\%) = 3.825\%$$

which matches the cumulative expectation $L_0 + \delta H_1 + \delta H_2 = 3.50\% + 0.15\% + 0.175\% = 3.825\%$.

6 Assumptions and Limitations

The probabilities produced by the framework are conditional on the following assumptions, which should be kept in mind when interpreting its output.

Support restriction. Futures prices identify only the mean of the post-meeting rate. Restricting the support to two adjacent steps gives, by Proposition 1, the minimum-dispersion distribution consistent with that mean, but it cannot distinguish, for instance, a 50/50 split between a hold and a cut from other distributions with the same mean, nor can it detect bimodal scenarios such as a hold versus a 50 bp cut. The method does not remove the need for a volatility model; it sidesteps it by choosing the least dispersed distribution.

Independence across meetings. The convolution in Eq. (23) applies the same increment distribution q_j at every state of the tree, which implies that the decision at meeting j is independent of the path followed up to meeting $j-1$. In practice, the Federal Reserve tends to move in sequences. The resulting cumulative probabilities are marginal and not path-conditional.

Reconciliation of redundant estimates. In a meeting month followed by a month without a meeting, $R_{\text{post},m}$ is identified twice: by the own-month formula and by the next-month anchor. The model uses the next-month contract when $N_m - M_m < L$ and the own-month formula otherwise. The gap between the two estimates is a useful diagnostic of the quality of the constant-rate assumption and of noise in the contracts.

Risk premium at long horizons. The zero-premium assumption is supported at horizons of one to three months but not at six to twelve months (Gürkaynak, Sack, and Swanson 2007; Piazzesi and Swanson 2008). Probabilities for meetings beyond approximately six months embed an unmodeled premium and should be interpreted as indicative.

Policy-rate proxy and unscheduled moves. The EFFR–floor spread is assumed to be constant (Section 4.6), so shifts in money market conditions are misread as policy expectations.

7 Conclusion

This research has set out a transparent framework for extracting market-implied probabilities of Federal Reserve policy decisions from 30-Day Federal Funds futures.

The day-weighted extraction equation isolates the expected post-meeting rate. Its multiplier makes the estimate fragile for meetings late in the month, which a next-month rule resolves. A two-point distribution on adjacent policy steps, shown to be the minimum-dispersion distribution consistent with the futures-implied mean, translates the expected move into probabilities, and forward convolution chains the meetings into a distribution over the policy path.

The framework is simple, replicable, and consistent with industry practice, but it identifies only the first moment of expectations and relies on a zero term premium, independence across meetings, and a stable EFFR-floor spread. Extensions that incorporate option-implied information, path dependence between meetings, and an explicit term-premium adjustment are natural directions for future work.

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Appendix A: Numerical Illustration

Setup. Assume that the pre-meeting rate is $R_{\text{pre},1} = 3.58\%$ (a lower bound of 3.50% and an EFFR–floor spread of 8 bp), and that the meetings of October 28, December 9, and January 28 are evaluated. The October and January meetings use the next-month estimator (Table 1), and the December meeting uses the own-month estimator. The assumed settlement prices are $P_{\text{Nov}} = 96.4700$, $P_{\text{Dec}} = 96.5500$, and $P_{\text{Feb}} = 96.6575$.

Table A1. *Extraction of expected post-meeting rates and meeting-level probabilities (hypothetical inputs).*

Meeting	Estimator	$R_{\text{pre},m}$ (%)	$R_{\text{post},m}$ (%)	H_m	α_m	Hold	Cut 25 bp
Oct. 28	Next-month (Nov.)	3.580	3.530	−0.200	0.200	80.0%	20.0%
Dec. 9	Own-month	3.530	3.417	−0.451	0.451	54.9%	45.1%
Jan. 28	Next-month (Feb.)	3.417	3.343	−0.299	0.299	70.1%	29.9%

Notes: For October, $R_{\text{post},m} = 100 - P_{\text{Nov}} = 3.530$. For December, Eq. (7) gives $R_{\text{post},m} = (31 \times 3.450 - 9 \times 3.530)/22 = 3.417$. For January, $R_{\text{post},m} = 100 - P_{\text{Feb}} = 3.3425$.

Error amplification. Had the October meeting been identified from its own-month contract, the implied average would have been $R_{\text{avg},m} = (28 \times 3.58 + 3 \times 3.53)/31 = 3.5752\%$ (price 96.4248). A one-basis-point error in this price would move the estimated post-meeting rate by 10.3 bp, shifting the cut probability from 20% to roughly 61%. With the next-month contract, the same one-basis-point error moves the post-meeting rate by just one basis point and the cut probability by four percentage points.

Convolution. Chaining the three meetings with Eq. (23) yields the cumulative target range distribution in Figure 3. After the third meeting, the model assigns 30.8% to no change, 46.1% to a cumulative 25 bp cut, 20.4% to 50 bp, and 2.7% to 75 bp. The mean cumulative change is -0.95 steps (-23.8 bp), equal to the sum of the H_m values, as stated in Eq. (24).

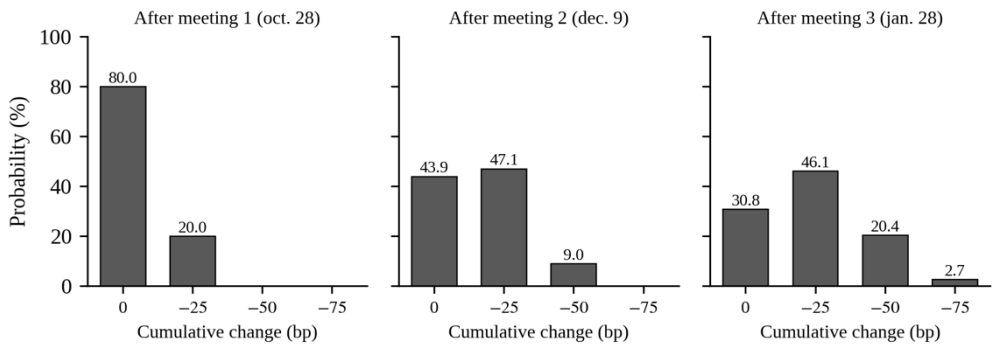


Figure 3. *Distribution of the cumulative change in the target range after each meeting (hypothetical inputs). Bars show probabilities in percent.*